

INTERNSHIP PROJECT REPORT

Data Cleaning and Exploratory Data Analysis Report

Netflix Movies & TV Shows Dataset

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Python • Pandas • NumPy • Matplotlib • Seaborn

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01 Introduction

Data cleaning is an essential step in the data analysis process. Real-world datasets often contain missing values, duplicate records, inconsistent data, and outliers that can affect the quality of analysis.

The objective of this project is to clean the Netflix Movies and TV Shows dataset, perform exploratory data analysis (EDA), visualize important trends, and derive meaningful insights using Python.

02 Objective

The objectives of this project are:

- Import and understand the dataset
- Identify missing values
- Handle missing data
- Detect and remove duplicate records
- Analyze outliers
- Perform exploratory data analysis
- Create informative visualizations
- Generate insights from the dataset
- Export the cleaned dataset

03 Dataset Description

The Netflix Movies and TV Shows dataset contains information about movies and television shows available on Netflix. The dataset includes the following attributes:

Column	Description
show_id	Unique identifier for each title
type	Movie or TV Show
title	Name of the content
director	Director name
cast	Cast members
country	Country of production
date_added	Date added to Netflix
release_year	Year of release
rating	Content rating
duration	Movie duration or TV show seasons
listed_in	Genre / category
description	Brief description

04 Software and Libraries Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Jupyter Notebook / VS Code

05 Data Cleaning Process

5.1 Loading the Dataset

The dataset was imported into Python using the Pandas library.

```
df = pd.read_csv("dataset.csv")
```

5.2 Understanding the Dataset

The following functions were used:

- head()
- shape
- info()
- describe()
- columns

These functions provided information about the number of rows, columns, data types, and statistical summaries.

5.3 Missing Value Treatment

Missing values were found in the following columns:

- Director
- Cast
- Country
- Date Added
- Rating

The missing values were handled using suitable methods:

Column	Treatment Applied
Director	"Unknown"
Country	"Unknown"
Cast	"Not Available"
Rating	Mode (most frequent value)
Date Added	"Unknown"

5.4 Duplicate Records

Duplicate records were detected using:

```
df.duplicated().sum()
```

These duplicate rows were removed using:

```
df.drop_duplicates(inplace=True)
```

5.5 Outlier Detection

Outlier analysis was performed using the Interquartile Range (IQR) method on the release_year column. A box plot was created to visualize the distribution of release years. Since older movies are genuine records rather than data-entry errors, they were analyzed carefully before deciding whether to remove them.

06 Exploratory Data Analysis

Several visualizations were created to understand the dataset:

- Movies vs TV Shows
- Distribution of Ratings
- Release Year Distribution
- Top Countries
- Top Directors
- Most Popular Genres
- Pie Chart of Content Types
- Box Plot of Release Years

07 Visualizations

The following graphs were created:

- Count Plot
- Histogram
- Bar Chart
- Pie Chart
- Box Plot

These visualizations helped identify important trends and patterns in the Netflix dataset.

08 Key Findings

Majority
MOVIES

USA
TOP COUNTRY

TV-MA
TOP RATING

Post-2000
GROWTH

The analysis revealed several important insights:

- The dataset contains both Movies and TV Shows.
- Movies make up the majority of Netflix content.
- Missing values were successfully handled.
- Duplicate records were removed.
- The United States contributes the largest number of Netflix titles.
- Drama and International Movies are among the most common genres.
- TV-MA is one of the most frequently occurring content ratings.
- A significant increase in content production is observed after the year 2000.
- Outlier analysis indicated that very old release years represent genuine historical content rather than incorrect data.

09 Challenges Faced

During the analysis, the following challenges were encountered:

- Handling missing values in multiple columns.
- Working with categorical features containing many unique values.
- Deciding whether to remove outliers that represented valid historical data.

These challenges were addressed using appropriate data-cleaning techniques and careful analysis.

10 Conclusion

This project successfully demonstrated the complete workflow of data cleaning and exploratory data analysis using Python. The dataset was cleaned by handling missing values, removing duplicate records, and analyzing outliers. Various visualizations were created to understand the distribution of Netflix content, production trends, genres, ratings, and countries.

The cleaned dataset is now suitable for further analysis and machine learning applications.

11 Future Scope

The project can be extended by:

- Building a recommendation system.
- Predicting content ratings using machine learning.
- Creating an interactive dashboard using Power BI or Tableau.
- Performing sentiment analysis on movie descriptions.
- Deploying the analysis as a web application.

12 References

- Netflix Movies and TV Shows Dataset
- Python Documentation
- Pandas Documentation
- Matplotlib Documentation
- Seaborn Documentation

13 Appendix

Libraries Used

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Output Files

- dataset.csv
- cleaned_netflix_dataset.csv
- netflix_analysis.ipynb
- README.md
- Data_Cleaning_Report.pdf